FUN-1.A.1 If f is a continuous function on the closed interval $[a, b]$ and d is a number between $f(a)$ and $f(b)$, then the Intermediate Value Theorem guarantees that there is at least one number c between a and b, such that $f(c) = d$.

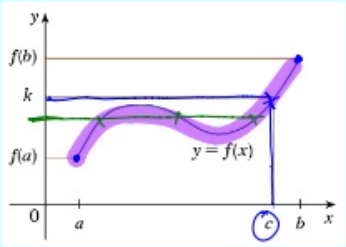
1-16 Intermediate Value Theorem

Suppose you were 4' 6" tall when you 10 years old and now you are 5' 7". Was there a time when you must have been exactly 5' tall? When would that time have occurred?

This is the concept behind the Intermediate Value Theorem.

Intermediate Value Theorem (IVT)	**If** f is continuous on the closed interval $[a, b]$, where $f(a) \neq f(b)$ and k is a number between $f(a)$ and $f(b)$, **then** there is at least one number c in $[a, b]$ such that $f(c) = k$

The IVT is about showing a point exists **<u>between</u>** two values.



Important Notes about the Intermediate Value Theorem (IVT)	1. The IVT is called an **existence** theorem. We will learn 3 of these theorems in AP Calculus. 2. When using a theorem, we must first show that the **conditions** are met. Condition 1: The function MUST be **continuous** on the given interval. Condition 2: The (y) value we want to guarantee MUST be **between** the (y) values at the endpoints. 3. AP problems rarely explicitly ask you to use the IVT…you have to recognize it by the wording! *****Must there be…** **Can you guarantee…** **Show that there must…** **Does a value exist…**

How to justify problems using the Intermediate Value Theorem (IVT)
1. Find the **function values** on the given interval by plugging values into the given function. 2. Show that the requested y value is **between** the two y values at the endpoints of the interval. 3. Make sure the function is **continuous** on the closed interval. **4. Invoke the IVT by using the following conclusion:** Since $f(x)$ is continuous on the interval _____ , where ___ ≠ ___ and ___ is a number between ___ and ___ , the IVT guarantees that … (reword the question as a statement)

x	1	3	4	7
$f(x)$	-2	1	4	6

Example 1: The function $f(x)$ is continuous with selected values of $f(x)$ shown in the table above. Is there a value c with $1 < c < 7$ such that $f(c) = 0$? Justify your answer.

Yes, since f is continuous on the closed interval $[1, 7]$, where $f(1) \neq f(7)$ and 0 is a number between $f(1) = -2$ and $f(7) = 6$, then the IVT guarantees there is at least one number c in $(1, 7)$ such that $f(c) = 0$.

Example 2: Use the Intermediate Value Theorem to show that $g(x) = x^3 + 2x - 1$ has a zero in the interval $[0,1]$.

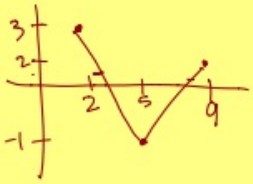
$g(0) = -1$ $g(1) = 2$

$f(c) = 0$
$0 \le c \le 1$

g is continuous on the closed interval $[0, 1]$, $g(0) = -1$ and $g(1) = 2$. Since $g(0) \neq g(1)$ and 0 is a number between $g(0) = -1$ and $g(1) = 2$, then the IVT guarantees there is at least one number c in $(0, 1)$ such that $g(c) = 0$.

Example 3: The functions $h(x)$ and $k(x)$ are continuous. Selected values for h and k are shown in the table below.

x	1	2	5	9
$h(x)$	5	3	-1	2
$k(x)$	-2	1	7	6



(a) On the interval $2 \leq x \leq 9$, what is the fewest number of times $h(x) = 1$?

Since h is continuous on the closed interval $[2, 9]$, by the IVT there are at least 2 times where $h(x) = 1$. In $[2, 5]$ and $[5, 9]$.

(b) Let m be the function defined as $m(x) = k(h(x))$. Can you guarantee a value c, where $1 < c < 9$ such that $m(c) = 4$? Give a reason for your answer.

$m(1) = k(h(1)) = k(5) = 7$
$m(9) = k(h(9)) = k(2) = 1$

Yes since $m(x)$ is continuous on the interval $[1,9]$ and $m(1) = 7$ and $m(9) = 1$, then the IVT guarantees there is a value c such that $m(c) = 4$ in the interval $[1,9]$ because 4 is between $m(1)$ and $m(9)$.